CHAPTER THREE: WEBSITE INTERFACE DESIGN



Learning Objectives:

By the end of this chapter the learner shall be able to;

i. design a web page ii. explain the website design basics iii. explain the principle of user interface design iv. explain the key elements of a good website

3.1 Introduction Web Interface Design

Website interface design involves making decisions on the look and feel of the website. This starts when the web developer collects the site requirements from the users. Using the defined user needs the website developer is able to decide which design is most appropriate for a certain website. User interface is the space where interaction between humans and website occurs. The goal of interaction between a human and a website at the user interface is effective operation and control of the website and feedback from the website which aids the user in make certain decisions. User interfaces provide a means of:

- Input, allowing the users to manipulate a website
- Output, allowing the website to indicate the effects of the users' manipulation

Web design is the process of designing websites - a collection of online content including documents and applications that reside on web servers. The process of web design includes planning, postproduction, research, advertising, as well as media control that is applied to the pages within the site by the designer or group of designers with a specific purpose. The site itself can be divided into its main page, also known as the home page, which cites the main objective as well as highlights of the site's daily updates; which also contains hyperlinks that functions to direct viewers to a designated page within the site's domain.

Website Design Basics

Following are some basic steps to be followed while creating web pages:

Create a Welcom/Index/first page of your web site.

- Avoid use of heavy graphics in the first page from the user_s point of view.
- Page should be descriptive and interactive both as per requirements.

Some minimum technical requirements might include:

- Fast loading of pages
- Presentation with clarity and readability - with or without graphics
- Validation of the contents
- Easy and clear navigation
- Instruction on how to use the site.

Identify and address all potential problems such as:

- Slow connections
- Physical constraints of the user such as sight or hearing impaired visitors
- Platform support
- Browser support

Web page guidelines: Design Considerations

- Follow a simple and consistent design. Complex designs can confuse people, so keep it simple. A consistent design will let your readers concentrate on content, without having to waste time figuring out how to maneuver your layout. At Cornell, the time and workstation "power" required to load a Web page are still important considerations when using some of the latest Web authoring tools, especially if your visitor is connecting to campus with a modem.

- Don't create gratuitous graphics. Graphics are one reason for the interest in the Web by both publishers and readers and should definitely be included on your Web page if possible. Still, it's important not to overuse them. Blinking text and other excessive decorations can be distracting; background colors and textures can affect download time. Think very carefully about the colors and textures you choose for backgrounds and the effects they'll have on the readability of text. Also, don't forget that many people still use monochrome monitors.

- Give people cross links. Visitors should be able to move from one major page to another on your site without having to go back to your home page. Put cross links to all your major pages at the bottom of all major pages. For a good example, see the bottom of this page.

- Be careful about "over-linking." While linking to the work of others in your organization and throughout the world can help your readers, it can also lead to information overload. It's important to balance linking within your page design. Too many links can be a visual eyesore on the page and a distraction from the original information you want your readers to concentrate on.

- Don't create dead end links. Readers can get discouraged from returning to your pages when those pages are filled with empty links with grand labels like Descriptions of All Classes!

3.2 Principles of User Interface Design

Principles of User Interface Design are intended to improve the quality of user interface design:

- ***The structure principle***: Design should organize the user interface purposefully, in meaningful and useful ways based on clear, consistent models that are apparent and recognizable to users, putting related things together and separating unrelated things, differentiating dissimilar things and making similar things resemble one another. The structure principle is concerned with overall user interface architecture.

- ***The simplicity principle***: The design should make simple, common tasks easy, communicating clearly and simply in the user's own language, and providing good shortcuts that are meaningfully related to longer procedures.

- ***The visibility principle***: The design should make all needed options and materials for a given task visible without distracting the user with extraneous or redundant information. Good designs don't overwhelm users with alternatives or confuse with unneeded information.

- ***The feedback principle***: The design should keep users informed of actions or interpretations, changes of state or condition, and errors or exceptions that are relevant and of interest to the user through clear, concise, and unambiguous language familiar to users.

- ***The tolerance principle***: The design should be flexible and tolerant, reducing the cost of mistakes and misuse by allowing undoing and redoing, while also preventing errors wherever possible by tolerating varied inputs and sequences and by interpreting all reasonable actions.

- ***The reuse principle***: The design should reuse internal and external components and behaviors, maintaining consistency with purpose rather than merely arbitrary consistency, thus reducing the need for users to rethink and remember.

3.3 Web usability

Web usability is an approach to make web sites easy to use for an end-user, without the requirement that any specialized training be undertaken. The user should be able to intuitively relate the actions he needs to perform on the web page, with other interactions he sees in the general domain of life e.g. press of a button leads to some action. The broad goal of usability can be:

- Present the information to the user in a clear and concise way.
- To give the correct choices to the users, in a very obvious way.
- To remove any ambiguity regarding the consequences of an action e.g. clicking on delete/remove/purchase.
- Put the most important thing in the right place on a web page or a web application.

3.4 Key Elements of a Good Website

The objective of any website is to get viewed and read as much as possible. Elements such as content, appearance and usability are some of the factors that set apart a great website from a good one. There are several factors that go into making a successful website. Besides its use, the appearance and userfriendliness are the most important factors that help make a website popular. Following are the key elements of a good website:

- Content - The most important element in any website is good content. Your content is the reason users visit your site. If it is superficial or badly written, you can be sure that they will leave fast and won't come back again. The copy for your website should be well framed and helpful Truthful information about your company, products, services, etc is a must. Information one must not forget to include is an 'about' page as well as contact information. The important aspect to keep in mind here is to provide information that is going to be useful to your user. Also, ensure that the content on your website is updated and does not contain any grammatical or spelling mistakes, for it is a direct reflection on your company's image.
- Visual Appeal - We all know what they say about first impressions, and this school of though extends to websites too. The decision to go through the content of a website is likely to be based

on how a visitor responds to its appearance. Your website design should be clean and simple, but all the same - eye-catching. A tasteful website isn't flashy. Blinking letters and flashing pictures are disturbing, and will turn away most readers instead of attracting them. They also make it a lot harder to read the information provided, which can be quite irritating for the user.

- User Interface - A good user interface is another key element of a good website. The user interface should be the kind an average user will find easy to use, and enable him or her to find whatever they want with ease. Part of this includes well-planned major (top level) categories, navigation that is easy to spot and consistent throughout the website. One way to achieve this is to have links to the key areas of your site in the top half of your website. This layout should remain consistent throughout the various pages. A good user interface will offer the same options in the footer, or even a few extra links. Larger websites with lots of pages should have a site map.

- Content Layout - A big mistake is to place text over backgrounds, for it makes reading difficult. Dark background colors do not make reading any easier. Studies have shown that black text on a white background is the best combination for reading and remembering. Large blocks of text should be split up into paragraphs containing 4-5 sentences. Ensure that your page contains sufficient white space, for clarity and neatness. Every page should have the same margins. There should be consistency throughout the website, so only one set of fonts, colors and layout should be used on all pages. On element that annoys most users is multiple scroll bars. Keep them at a minimum and try to avoid the side-to-side scroll bars.

- Search Tool - Good websites containing many pages have a search field to assist their readers. This feature is very helpful, and enables potential customers to find whatever they want with ease. One can make use of use a Google Search on their site, or WordPress (or another blogging platform or CMS / Content Management System). It will be a little more challenging with a static html site, but there are services available that will let one incorporate a functional search box onto their site.

- Separate Design from Content - The best developed sites on the Internet combine the use of XHTML and CSS (Cascading Style Sheets), which create a separation of design vs content. Among its many advantages are that one has to update just one CSS file to see the change implemented site-wide. Also this separation of content from design will result in quick loading (search engines will no longer have to go through excess code to find out if the content is relevant). On the same note, minimize your utilization of flash animations, music, or video or

-

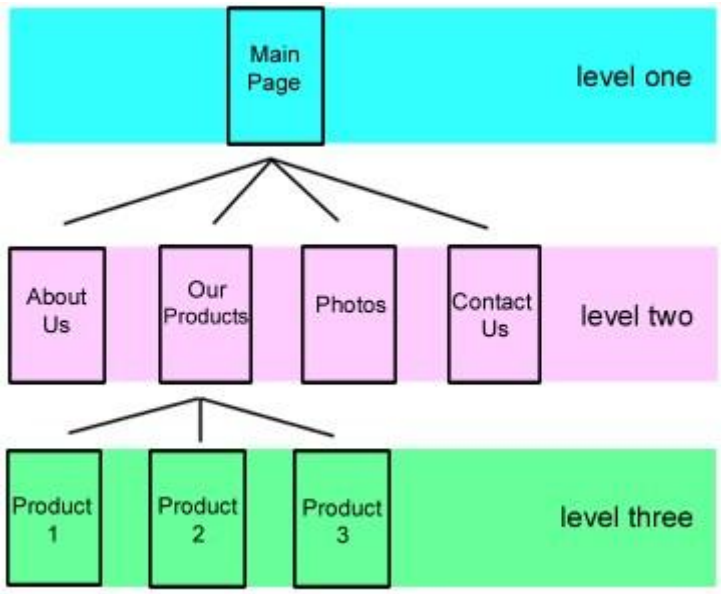
  anything that is going to make your site take a long time to load. Web surfers have very little patience and a short attention span.

  Cross Browser Compatibility - Another key element of a good website is that it displays well in as many of the mainstream web browsers as possible and is also compatible across platforms. Although most Internet users utilize Firefox, you don't want to leave out those using Internet Explorer and the other few browsers. Also, the success of your website can be hinged on how well it is optimized for search engines.

- Web Optimized Images - Good picture quality on a website instantly leaves a user impressed. Most people save all their images in a compressed format. So be careful while doing this as your images will appear pixelated when you upload them on your site.

3.5 Web Site Organization

A well organized web site will increase its usability resulting in your visitors staying on your site longer and coming back more frequently. The main page of your site, also referred to as the home or index page, is the first page visitors will see when going to your domain. On this page there should be navigation linking to each of the major subpages of your site. These navigation links can be either text links or image buttons. For example, imagine a fictional company who wants a web site as a sort of online brochure. They have 3 products to sell, they don't want to sell them online, but would like detailed information about each of them to be available on their site. They also want an About Us, a Contact Us, and a Photos page. A products page could be set up that gives an introduction about their products, and then links to 3 separate pages each of which describes one of the products in detail. The main page could be referred to as level one, the Our Products, About Us, Contact Us, and the Photos page as level 2, and each of the individual product pages as level three. See diagram below.



There would then be five main navigational links for this site:

-

  Main page

- About Us
- Our Products
- Photos
- Contact Us

These five links should appear on each of the eight pages of the site. The header and title on the top of each page should either be the same as or similar to the text on each of the navigation links. In this way whichever page of the site a visitor finds themselves on, either through moving around the site on their own, or by entering the site on any of its eight pages through a search engine link, they should be able to easily figure out exactly where on the site they are.

To maintain a consistent look and feel throughout the site, the same overall design should be used on each page. For example, if each of the five main navigation links were made from buttons shaped like hats, and they were lined up along a left column, they should be that way on each page of the site. It is however, a good idea to include plain text links to each of the main pages on the bottom of each page of the site. This is especially important if the other links are made from graphics. There are people who use browsers with either no graphics capability, or they have them turned off. They will need a way to navigate through your site also.

It has also been stated that it should not take more than three clicks to get from one page to any other page on the same web site, and if it does, the site just isn't organized well.

3.6 Website wireframe

A website wireframe, also known as a page schematic or screen blueprint, is a visual guide that represents the skeletal framework of a website. The wireframe depicts the page layout or arrangement of the website‘s content, including interface elements and navigational systems, and how they work together. The wireframe usually lacks typographic style, color, or graphics, since the main focus lies in functionality, behavior, and priority of content. Website wireframe focuses on ―what a screen does, not what it looks like.‖ Wireframes focus on:

- The kinds of information displayed
- The range of functions available

- The relative priorities of the information and functions
  The rules for displaying certain kinds of information
- The effect of different scenarios on the display

The website wireframe connects the underlying conceptual structure, or information architecture, to the surface, or visual design of the website. Wireframes help establish functionality, and the relationships between different screen templates of a website. An iterative process, creating wireframes is an effective way to make rapid prototypes of pages, while measuring the practicality of a design concept. Wireframing typically begins between ―high-level structural work—like flowcharts or site maps—and screen designs. Within the process of building a website, wireframing is where thinking becomes tangible.

Uses of wireframes

Developers use wireframes to get a more tangible grasp of the site‘s functionality, while designers use them to push the user interface (UI) process. User experience designers and information architects use wireframes to show navigation paths between pages. Business stakeholders use wireframes to ensure that requirements and objectives are met through the design.

Working with wireframes may be a collaborative effort since it bridges the information architecture to the visual design. Due to overlaps in these professional roles, conflicts may occur, making wireframing a controversial part of the design process. Since wireframes signify a ―bare bones‖ aesthetic, it is difficult for designers to assess how closely the wireframe needs to depict actual screen layouts. Another difficulty with wireframes is that they don‘t effectively display interactive details. Modern UI design incorporates various devices such as expanding panels, hover effects, and carousels that pose a challenge for 2-D diagrams. Wireframes may have multiple levels of detail and can be broken up into two categories in terms of fidelity, or how closely they resemble the end product.

- Low-fidelity Resembling a rough sketch or a quick mock-up, low-fidelity wireframes have less detail and are quick to produce. These wireframes help a project team collaborate more effectively since they are more abstract, using rectangles and labeling to represent content.
- High-fidelity High-fidelity wireframes are often used for documenting because they incorporate a level of detail that more closely matches the design of the actual webpage, thus taking longer to create.

Elements of wireframes

The skeleton plan of a website can be broken down into three components: information design, navigation design, and interface design. Page layout is where these components come together, while wireframing is what depicts the relationship between these components.

- Information Design - Information design is the presentation—placement and prioritization of information in a way that facilitates understanding. Information design is an area of graphic design, meant to display information effectively for clear communication. For websites, information elements should be arranged in a way that reflects the goals and tasks of the user.

- Navigation Design - The navigation system provides a set of screen elements that allow the user to move page to page through a website. The navigation design should communicate the relationship between the links it contains so that users understand the options they have for navigating the site. Often, websites contain multiple navigation systems such as a global navigation, local navigation, supplementary navigation, contextual navigation, and courtesy navigation.

- Interface Design - User interface design includes selecting and arranging interface elements to enable users to interact with the functionality of the system. The goal is to facilitate usability and efficiency as much as possible. Common elements found in interface design are action buttons, text fields, check boxes, radio buttons and drop-down menus.

Chapter Review Questions

1. Can you give the minimum technical requirements that you can outline for a new site
2. outline the principles of user interface design
3. outline the key elements of a good website
4. define a website wireframe
5. what are the elements of a wireframe

Suggested Further Reading

1. Niederst J., Learning Web design, Shroff Publishers

2. Peter, K., Web design, Tools and Techniques, Peach Pit press

3. http://www.w3schools.com